



YEAR 12

COMPUTER SCIENCE

Semester 2 Examination, 2024

Examination Solutions

Section One: Short answer**40% (93 marks)****Question 1****(6 marks)**

Complete the code below for a Python function named *count_vowels* that takes a string (*input_string*) as a parameter and returns the number of vowels in that string. Ensure the function is case-insensitive.

Description	Mark
Initialises a counter variable to zero	1
Use of iteration to go through each letter of the <i>input_string</i>	1
Converts the input string to lowercase to ensure the function is case-insensitive	1
Use of selection to check current character is a vowel	1
If it is, it increments the count by one	1
Returns the total count of vowels	1
Total	6
Sample answer: def count_vowels(input_string): vowels = 'aeiou' count = 0 for char in input_string.lower(): if char in vowels: count += 1 return count	
Accept other relevant answers	

Question 2**(3 marks)**

Describe the concept of polymorphism in object-oriented programming and outline **one** benefit polymorphism offers to developers.

Description	Mark
Describes the concept of polymorphism in object-oriented programming	2
Provides relevant facts about the concept of polymorphism in object-oriented programming	1
Subtotal	2
Outlines a benefit of polymorphism to developers	1
Subtotal	1
Total	3
Sample answer: Polymorphism: Polymorphism is a fundamental concept in object-oriented programming that allows objects of different classes to be treated as objects of a common superclass. It is implemented through method overriding and interfaces, enabling a single interface to represent different underlying data types. Benefit: Answers could include: • Code Reusability: Polymorphism enables writing more generic and reusable code. • Simplified Code: Reduces complexity and enhances readability by handling different data types with a single interface. • Flexibility: Allows easy extension of code by adding new classes with minimal changes to existing code.	
Accept other relevant answers	

Question 3**(4 marks)**

Outline each of the following ACID properties in database transactions.

Description	Mark
For each of the four properties (4 x 1 mark)	
Outlines each ACID property concisely	1
Total	4
Sample answer: Atomicity: Ensures that all operations in a transaction are completed; if not, the transaction is aborted. Consistency: Guarantees that a transaction brings the database from one valid state to another. Isolation: Ensures that transactions are executed in isolation, meaning changes made in one transaction are not visible to other transactions until the transaction is committed. Durability: Ensures that once a transaction is committed, it remains so, even in the event of a system failure.	
Accept other relevant answers	

Question 4**(4 marks)**

Describe how each of the following factors influence the quality of data being stored in a database.

Description	Mark
For each of the two factors (2 x 2 marks)	
Describes how the factor influences the quality of data being stored	2
Provides relevant facts about how the factor influences the quality of data being stored	1
Total	4
Sample answers: Relevance influences data quality by ensuring that the data stored is pertinent to the specific needs and objectives of the organisation. Data that is relevant is useful for decision-making and analysis, while irrelevant data can lead to inefficiencies and misguided conclusions. Accuracy affects data quality by ensuring that the data correctly reflects the real-world entities or events it represents. Accurate data is essential for making reliable decisions and avoiding errors, whereas inaccurate data can result in errors, misinformed decisions, and reduced credibility.	
Accept other relevant answers	

Question 5

(5 marks)

- (a) Explain the role of the Transmission Control Protocol (TCP) in data transmission over a network. (3 marks)

Description	Mark
Explains the role of TCP in data transmission	3
Describes the role of TCP in data transmission	2
Provides relevant facts about TCP	1
Total	3
Sample answer: TCP is responsible for ensuring reliable data transmission by establishing a connection between the sender and receiver before data transfer begins. It uses mechanisms like error checking, data packet sequencing, and acknowledgments to guarantee that data is delivered accurately and ensures that packets are reassembled in the correct sequence at the receiving end.	
Accept other relevant answers	

- (b) Outline **two** ways network segmentation is beneficial to a corporate network environment. (2 marks)

Description	Mark
For each of the two properties (2 x 1 mark)	
Outlines a way in which network segmentation is beneficial to a corporate network environment	1
Total	2
Answers could include: - Enhanced Security: Network segmentation limits the spread of cyber-attacks by isolating sensitive systems. - Improved Performance: It reduces network congestion by confining broadcast traffic to specific segments. - Simplified Compliance: Segmentation helps ensure compliance with regulatory requirements by isolating data for audits. - Easier Management: It simplifies network management by allowing specific policies to be applied to different segments.	
Accept other relevant answers	

Question 6**(2 marks)**

Your development team is tasked with creating a new mobile app for a client who wants regular updates and feedback incorporated into the design process.

Compare the iterative and linear development processes. Outline **two** factors that may affect your choice of which model to use for this project.

Description	Mark
For each of the two factors (2 x 1 mark)	
Outlines a factor that may affect the decision	1
Total	2
Sample answers could include: • Client involvement: The iterative process allows for regular updates and feedback, which suits clients who want continuous input. • Flexibility in design: Iterative development is more adaptable to changes and client feedback compared to the linear process.	
Accept other relevant answers	

Question 7**(5 marks)**

The Caesar cipher is one of the earliest and simplest methods of encryption. It involves shifting each letter in a plaintext message a fixed number of places down the alphabet. For example, with a shift of 3, 'A' would be encrypted to 'D', 'B' to 'E', and so on.

- (a) Describe **one** weakness of this method of encryption. (2 marks)

Description	Mark
Describes the weakness	2
Provides relevant facts about the weakness	1
Total	2
Sample answer: The Caesar cipher is vulnerable to frequency analysis, as it does not significantly alter the frequency distribution of letters, making it easy for attackers to decipher the encrypted message.	
Accept other relevant answers	

Current practices in encryption have evolved significantly, incorporating more sophisticated techniques such as symmetric and asymmetric encryption to enhance security.

- (b) Explain how asymmetric encryption can be used to authenticate data being sent across a network. (3 marks)

Description	Mark
Explains the use of asymmetric encryption for authentication	3
Describes the use of asymmetric encryption for authentication	2
Provides relevant facts about the use of asymmetric encryption for authentication	1
Total	3
Sample answer: Asymmetric encryption uses a pair of keys, a public key and a private key, to authenticate data. When a sender encrypts a message with their private key, the recipient can verify the sender's identity by decrypting it with the sender's public key, ensuring the message's authenticity.	
Accept other relevant answers	

Question 8**(4 marks)**

A software development team is working on a new application that incorporates several third-party libraries and open-source components. One of the developers discovers a useful piece of code in an online forum and decides to integrate it into the project.

- (a) Describe the importance of acknowledging the intellectual property (IP) of others in software development. (2 marks)

Description	Mark
Describes the importance	2
Provides relevant facts about the importance	1
Total	2
Sample answer: Acknowledging the intellectual property of others in software development is crucial because it respects the original creators' rights and contributions, fostering a culture of integrity and collaboration. It also helps to avoid legal issues and potential copyright infringements, which can lead to costly disputes and damage to reputation.	
Accept other relevant answers	

- (b) Outline **two** potential consequences for the development team if they fail to acknowledge the use of third-party code. (2 marks)

Description	Mark
For each of the two consequences (2 x 1 mark)	
Outlines a potential consequence	1
Total	2
Sample answers could include: • Legal Action: The team may face legal consequences, including lawsuits for copyright infringement, leading to financial penalties and project delays. • Reputation Damage: Failing to acknowledge third-party code can damage the team's and company's reputation, leading to loss of trust from clients and the developer community. • Project Delays: Legal disputes or the need to remove and replace uncredited code can cause significant project delays. • Financial Loss: Penalties, legal fees, and potential settlements can result in substantial financial losses. • Loss of Business Opportunities: Clients and partners may be reluctant to work with a team that does not respect intellectual property rights, resulting in lost business opportunities.	
Accept other relevant answers	

Question 9

(11 marks)

Given the following un-normalised table MusicLessons.

Booking ID	StudentName	Phone	LessonDate	Instrument	TeacherName	LessonFee	Duration
1	Emma Johnson	93652451	2023-04-01	Piano	Alfred Smith	50.00	1 hour
2	Liam Brown	94448568	2023-04-02	Guitar	Jane Dease	45.00	1 hour
3	Emma Johnson	93652451	2023-04-03	Piano	Alfred Smith	50.00	1 hour
4	Olivia White	92235151	2023-04-04	Violin	Paul James	55.00	1 hour
5	Liam Brown	94448568	2023-04-05	Clarinet	Jane Dease	90.00	2 hours

- (a) Outline each of the following anomalies and provide an example of each from the data in the MusicLessons table. (6 marks)

Description	Mark
For each of the three anomalies (3 x 2 marks)	
Outlines the anomaly	1
Uses an example from the data provided	1
Subtotal	2
Total	6
Sample answer: Insert Anomaly: - Description: Occurs when certain information cannot be inserted into the database without the presence of other data. - Example: If a new teacher named "Mary Ann" needs to be added but no lesson is scheduled with her yet, the current structure does not allow for the teacher to be added independently. Update Anomaly: - Description: Occurs when data redundancy leads to inconsistencies when updating data. - Example: If Liam Brown's phone number changes, every record with Liam Brown's details needs to be updated. If any record is missed, inconsistencies will arise. Delete Anomaly: - Description: Occurs when the deletion of data inadvertently leads to loss of additional data. - Example: If Liam Brown's lesson on 2023-04-05 is deleted, all information about Jane Dease as a clarinet teacher would also be lost, assuming no other clarinet lessons are associated with her.	
Accept other relevant answers	

(b) Normalise the data to third normal form (3NF).

(5 marks)

Description	Mark
1 mark for each correct relation (must include relation name, primary key, foreign key(s) and / or non-key fields)	1
Total	5
Sample answer: Student (<u>StudentID</u> , StudentFirstName, StudentSurname, Phone) Instrument (<u>InstrumentID</u> , Instrument) Teacher (<u>TeacherID</u> , TeacherFirstName, TeacherSurname) Teacher/Instrument (TeacherID FK, <u>InstrumentID</u> FK, LessonFee) Booking (<u>BookingID</u> , LessonDate, Duration, StudentID FK, InstrumentID FK, TeacherID FK)	
Accept other relevant answers	

Question 10**(4 marks)**

Describe the difference between a one-to-one relationship and a one-to-many relationship in a relational database and outline **one** example for each.

Description	Mark
Describes the difference between the cardinalities	2
Provides relevant facts about the difference between the cardinalities	1
Subtotal	2
For each of the two cardinalities (2 x 1 mark)	
Outlines a correct example for the cardinality	1
Subtotal	2
Total	4
Sample answer: A one-to-one relationship means each record in one table is linked to exactly one record in another table, whereas a one-to-many relationship means each record in one table can be linked to multiple records in another table. One-to-One Examples: • Each country can have one capital city. • Each employee has one assigned company car. • Each book has one ISBN. One-to-Many Examples: • A single customer can place multiple orders in an online shopping database. • A single author can write multiple books. • A single teacher can have multiple students in a class. Also accept ERDs drawn to reflect these relationships using crow's feet notation Accept other relevant answers	

Question 11**(4 marks)**

Define the term "zero-day vulnerability" and describe why it poses a significant risk to organisations.

Description	Mark
Defines the term zero-day vulnerability	2
Provides relevant facts about zero-day vulnerability	1
Subtotal	2
Describes reason for risk to organisations	2
Provides relevant facts about reason for risk to organisations	1
Subtotal	2
Total	4
Sample answer: Zero-day vulnerability: A zero-day vulnerability is a software security flaw that is unknown to the software vendor and for which no patch or fix is available. Risk to organisations: It poses a significant risk because attackers can exploit the vulnerability before the vendor has a chance to address it, leading to potential data breaches, system compromises, and financial losses.	
Accept other relevant answers	

Question 12**(4 marks)**

Describe the concept of version control in programming and describe why it is important in managing changes to project code.

Description	Mark
Describes the concept of version control	2
Provides relevant facts about version control	1
Subtotal	2
Describe why it is important in managing changes to project code.	2
Provides relevant facts about why it is important in managing changes to project code	1
Subtotal	2
Total	4
Sample answer: Version control: Version control is a system that records changes to files over time so that you can recall specific versions later, enabling multiple developers to work on the same project without overwriting each other's work. Importance: Version control is important because it allows teams to track and manage changes, facilitates collaboration, helps resolve conflicts, and provides a history of modifications for rollback if necessary.	
Accept other relevant answers	

Question 13**(3 marks)**

Explain the difference between a blue team and a red team when involved in penetration testing.

Description	Mark
Explains the difference between blue team and red team	3
Provides detailed roles and objectives of each team	2
Gives a brief overview of the roles and objectives of each team	1
Total	3
Sample answer: The blue team is responsible for defending the network in a penetration test while the red team is offensive and responsible for attacking the network, both working together to enhance overall cybersecurity. The blue team defends an organisation's information systems by maintaining security, monitoring for potential threats, and responding to incidents. They work to detect, mitigate, and prevent cyber-attacks. The red team simulates cyber-attacks by attempting to exploit vulnerabilities in the system to test the defences and identify weaknesses. Their goal is to improve the organisation's security posture by highlighting areas that need strengthening.	
Accept other relevant answers	

Question 14**(2 marks)**

Describe the purpose of using stubs in software development.

Description	Mark
Describes the purpose of using stubs	2
Provides superficial comment about the purpose of using stubs	1
Total	2
Sample answer: Stubs are used in software development to simulate the behaviour of complex modules that are not yet implemented, allowing developers to test other parts of the system without waiting for the full functionality to be available.	
Accept other relevant answers	

Question 15**(3 marks)**

Explain the concept of unit testing including the process and purpose.

Description	Mark
Explains the concept of unit testing including the process and purpose	3
Describes the concept of unit testing in including the process and purpose	2
Provides basic facts about the process and purpose of unit testing	1
Total	3
Sample answer: Unit testing involves validating individual components of software, such as functions, methods, or classes, to ensure they function as expected. The process includes writing test cases for specific elements like a sorting function, a method calculating a total price, or a class handling user authentication, running these tests to check for correctness, and identifying bugs early in the development cycle. The purpose is to enhance code quality, simplify debugging, and facilitate maintenance by ensuring that each part works independently before integration.	
Accept other relevant answers	

Question 16**(6 marks)**

Examine the following Python code that tracks a player's score in a game. For each of the variables below, identify the scope of the variable and outline how the scope of each variables affects their usage in the code.

```
player_score = 500
```

```
def complete_level(level_points):
    global player_score
    bonus_points = 100
    level_score = level_points + bonus_points
    player_score += level_score
    return level_score
```

```
level_points = 300
level_score = complete_level(level_points)
print(f"Score for this level: {level_score}")
print(f"Total player score: {player_score}")
```

Description	Mark
For each of the three variables (3 x 2 marks)	
Describes the scope of each variable	1
Outlines how the scope of each variable affects its usage	1
Total	6
Sample answer: player_score: • Scope: Global • Usage in code: player_score is defined outside of any function, making it a global variable. It is accessible throughout the entire code, including inside the complete_level function due to the global keyword. This allows the function to modify the player_score variable directly. bonus_points: • Scope: Local to complete_level function • Usage in code: bonus_points is defined inside the complete_level function, making it a local variable. It is only accessible within the complete_level function and is used to calculate the level_score. level_points: • Scope: Local to both the main code block and complete_level function (different instances) • Usage in code: The level_points variable has two instances with local scope: one is defined in the main code block and passed as an argument to the complete_level function. Inside the function, level_points is used to calculate level_score and update player_score.	
Accept other relevant answers	

Question 17**(3 marks)**

A company needs to ensure that its remote employees can securely access its network and resources.

Explain **one** solution that could be used by the company to allow secure remote access.

Description	Mark
Explains a solution for secure remote access using a virtual private network (VPN)	3
Provides some basic facts associated with the use of a virtual private network (VPN)	2
Identifies VPN	1
Total	3
Sample answer: A Virtual Private Network (VPN) is a solution that can be used by the company to allow secure remote access. A VPN creates an encrypted tunnel between the remote employee's device and the company's network, ensuring that all data transmitted over the internet is secure and private. This encryption prevents unauthorised access and protects sensitive information from being intercepted by hackers. By masking the user's IP address, a VPN also enhances privacy, allowing employees to safely access the company network and resources from any location.	
Accept other relevant answers	

Question 18**(4 marks)**

A university is deploying wireless access points throughout its campus for its students and staff.

Describe the role of a wireless access point in a network and outline **two** factors that can affect its performance and security.

Description	Mark
Describes the role about the role of a wireless access point	2
Provides superficial comment about the role of a wireless access point	1
Subtotal	2
For each of the two factors (2 x 1 mark)	
Outlines a factor that can affect performance	1
Subtotal	2
Total	4
Sample answer: A wireless access point (WAP) acts as a central transmitter and receiver of wireless radio signals, enabling wireless devices to connect to a wired network. It bridges the gap between wired and wireless segments, providing internet access and facilitating network communication between wireless devices within a network. Possible factors could include: • Interference: Physical obstructions and other electronic devices can cause interference, reducing signal strength and network performance. • Access point placement: Poor placement of access points can lead to dead zones or areas with weak signal coverage, affecting performance. • Network Congestion: High numbers of connected devices can lead to network congestion, reducing the performance and speed of the wireless network.	
Accept other relevant answers	

Question 19**(5 marks)**

A video streaming service provider needs to ensure that users experience minimal buffering and high-quality streaming.

- (a) Define the term 'bandwidth'. (2 marks)

Description	Mark
Defines the term 'bandwidth'	2
Provides superficial comment about the term bandwidth	1
Total	2
Sample answer: Bandwidth is the maximum rate at which data can be transferred over a network connection in a given amount of time, typically measured in bits per second (bps). It determines the capacity of the network to handle data traffic.	
Accept other relevant answers	

- (b) Discuss bandwidth's importance in network performance, particularly for video streaming services. (3 marks)

Description	Mark
Explains the importance of bandwidth for network performance for the video streaming service	3
Describes the importance of bandwidth in network performance for the video streaming service	2
Provides some relevant facts about the importance of bandwidth in network performance	1
Total	3
Sample answer: Bandwidth is crucial for network performance as it determines how much data can be transmitted in a given time frame. For video streaming services, high bandwidth is essential to ensure that large amounts of video data can be delivered quickly and efficiently, resulting in minimal buffering and high-quality streaming. Insufficient bandwidth can lead to interruptions, lower resolution, and a poor user experience. Thus, adequate bandwidth is necessary to handle the data load of multiple users streaming videos simultaneously.	
Accept other relevant answers	

Question 20**(3 marks)**

Explain the purpose of ports in network transmission.

Description	Mark
Explains the purpose of ports in network transmission	3
Describes the purpose of ports in network transmission	2
Provides some relevant facts about the purpose of ports in network transmission	1
Total	3
Sample answer: Ports serve as communication endpoints for different services or applications within a network. They allow a single network device to support multiple connections simultaneously by directing incoming and outgoing data to the correct application or service. Each port is associated with a specific number that identifies the type of service, such as HTTP (port 80) or HTTPS (port 443). This ensures that data packets are accurately delivered to the appropriate application, facilitating organised and efficient network communication.	
Accept other relevant answers	

Question 21**(2 marks)**

Describe the importance of validating a software solution against its design specifications.

Description	Mark
Provides a description of the importance of validating a software solution against its design specifications	2
Provides some relevant facts for the importance of validating a software solution against its design specifications	1
Total	2
Sample answer: Validating a software solution against its design specifications ensures that the final product meets the intended requirements and functions as expected. It helps identify and rectify any deviations or defects early, thereby ensuring reliability, reducing development costs, and increasing user satisfaction by delivering a product that fulfills the specified needs.	
Accept other relevant answers	

Question 22**(3 marks)**

A community health organisation relies heavily on information from the internet to educate the public about health and wellness through its website. Recently, there has been a surge of online articles and social media posts promoting a new "miracle cure" for a popular strain of influenza. However, this information is unsupported by scientific evidence and has not been verified by any reputable health authority.

Discuss the potential impacts on public health if the community health organisation disseminates the unsupported and unverifiable information about the "miracle cure."

Description	Mark
Provides a detailed discussion of potential impacts	3
Discusses several impacts with some detail	2
Provides a basic discussion of potential impacts	1
Total	3
Sample answer: Disseminating unsupported and unverifiable information about the "miracle cure" can have serious negative impacts on public health. Firstly, it may lead individuals to forego proven medical treatments in favour of the unverified cure, potentially worsening their condition. Secondly, it can erode public trust in the health organisation and in scientifically validated medical advice, making it harder to communicate effective health interventions in the future. Lastly, it can contribute to the spread of misinformation, causing confusion and panic, and undermining efforts to control the influenza outbreak effectively.	
Accept other relevant answers	

Question 23**(3 marks)**

Explain the importance of using strong passwords for network security.

Description	Mark
Explains the importance of strong passwords for network security	3
Discusses the importance with some detail	2
Provides some basic facts about the importance	1
Total	3
Sample answer: Strong passwords are crucial in safeguarding sensitive information and preventing unauthorised access to network resources. They significantly reduce the risk of password guessing or brute-force attacks, thereby enhancing the overall security of the network. Weak passwords can lead to data breaches and unauthorised access, compromising network integrity and resulting in financial loss and reputational damage.	
Accept other relevant answers	

End of Section One

See next page

Section Two: Extended answer

60% (129 marks)

Question 24

(38 marks)

- (a) Use the information provided to develop the *Spacecraft* class in Python with the attributes for the spacecraft's name, type, a list of equipment onboard and the maximum weight of equipment the spacecraft can handle.

Include a method to display the spacecraft's details. Ensure the data is dynamically displayed as per the image shown above based on the sample data provided.

(13 marks)

Description	Mark
<code>__init__</code> method initialised	1
Use of appropriate parameters for <code>__init__</code> method - self, name, spacecraft_type, equipment, max_weight	1
Assignment of parameters to attributes – <code>self.name = name</code> <code>self.spacecraft_type = spacecraft_type</code> <code>self.equipment = equipment</code> <code>self.max_weight = max_weight</code>	1 1 1 1
<code>display_details</code> method initialised including use of self parameter	1
Correct output to print class attributes <code>print(f"Spacecraft Name: {self.name}")</code> <code>print(f"Spacecraft Type: {self.spacecraft_type}")</code> <code>print(f"Maximum weight: {self.max_weight}kg")</code>	1 1 1
Output of equipment <code>print("Equipment List:")</code> <code>for item in self.equipment:</code> <code>print(f" - {item}")</code>	1 1 1
Total	13

Sample answer:

class Spacecraft:

```
def __init__(self, name, spacecraft_type, equipment, max_weight):
    self.name = name
    self.spacecraft_type = spacecraft_type
    self.equipment = equipment
    self.max_weight = max_weight
```

```
def display_details(self):
    print(f"Spacecraft Name: {self.name}")
    print(f"Spacecraft Type: {self.spacecraft_type}")
    print(f"Maximum weight: {self.max_weight}kg")
    print("Equipment List:")
```

```
for item in self.equipment:  
    print(f" - {item}")
```

Accept other relevant answers that perform the desired function

- (b) SEA spacecraft are only able to carry a limited amount of weight for each mission. Therefore, it is essential to calculate the total weight of the equipment before each mission.

The weights of all the types of equipment are saved in a dictionary.

```
equipment_weights = {
    "Oxygen Tanks": 150,
    "Food Supplies": 200,
    "Fuel Cells": 300,
    "Medical Supplies": 100,
    "Scientific Instruments": 250
}
```

- (i) Develop a standalone function *calculate_total_weight* that takes a one-dimensional array of the equipment intended to be taken on the spacecraft (*equipment_list*) and the dictionary of the weights of the equipment (*equipment_weights*) and calculates and returns the total weight. (7 marks)

Description	Mark
Function <i>calculate_total_weight</i> defined	1
Correct use of parameters <i>equipment_list</i> , <i>equipment_weights</i>	1
Initialises <i>total_weight</i> to 0	1
Iterates through <i>equipment_list</i>	1
Correctly accesses weights from <i>equipment_weights</i>	1
Accumulates total weight	1
Returns the correct total weight	1
Total	7
Sample answer: <pre>def calculate_total_weight(equipment_list, equipment_weights): total_weight = 0 for item in equipment_list: total_weight += equipment_weights[item] return total_weight</pre>	
Accept other relevant answers that perform the desired function	

- (ii) Write Python code below to demonstrate the usage of the *calculate_total_weight* function. Provide the code to calculate the weights and to print the total weight. (4 marks)

Equipment: Oxygen Tanks, Fuel Cells, Medical Supplies

Description	Mark
Define equipment_list with the specified items	1
Call calculate_total_weight with equipment_list and equipment_weights	1
Store the result in total_weight	1
Print the total_weight	1
Total	4
Sample answer: <pre>equipment_list = ["Oxygen Tanks", "Fuel Cells", "Medical Supplies"] total_weight = calculate_total_weight(equipment_list, equipment_weights) print(f"Total weight: {total_weight}")</pre>	
Accept other relevant answers that perform the desired function	

It's important to ensure that the weight of the equipment to be carried on the spacecraft does not exceed the maximum weight that they can carry.

- (iii) Write Python code below to check for this requirement and to print an error message in the case that the equipment weight exceeds the capacity of the spacecraft. (2 marks)

Description	Mark
Conditional check for weight exceeding maximum capacity	1
Print error message if weight exceeds capacity	1
Total	2
Sample answer: <pre>if total_weight > spacecraft.max_weight: (1) print("Warning: Total weight exceeds the maximum capacity of the spacecraft!") (1)</pre>	
Accept other relevant answers that perform the desired function	

The Space Exploration Agency (SEA) is planning to expand the new software system for use in managing its spacecraft inventories, crew assignments, and mission schedules. As a developer, you are responsible for ensuring that the software adheres to ethical and legal standards.

- (c) State **two** ethical considerations you must consider when developing this software and, for each, describe **one** example of how the consideration could impact the SEA if not properly addressed. (6 marks)

Description	Mark
For each of the two ethical considerations (2 x 1 mark)	
States an ethical consideration	1
Subtotal	2
For each of the two ethical considerations (2 x 1 mark)	
Describes an example of potential impact	2
Provides basic superficial facts about the potential impact	1
Subtotal	4
Total	6
Sample answers could include: • Data Privacy - If the software does not adequately protect sensitive data, such as personal information of crew members, it could lead to data breaches. This could result in legal penalties, loss of trust, and damage to SEA's reputation. • Accuracy and Reliability - If the software contains errors or inaccuracies in managing spacecraft inventories, crew assignments, or mission schedules, it could lead to mission failures, endangering the lives of crew members and causing financial and operational losses for SEA.	
Accept other relevant answers	

As part of the SEA's new software system, you are tasked with ensuring that the software development follows good programming practices.

- (d) Describe the importance of validating input before processing in software development. (2 marks)

Description	Mark
Describe the importance of validating input before processing in software development	2
Provides relevant details about the importance of validating input before processing in software development	1
Total	2
Sample scenarios could include: Validating input before processing ensures that the data entered into the system is correct and meets the required criteria. This helps prevent errors, security vulnerabilities, and unexpected behaviour in the software.	
Accept other relevant answers	

SEA's new software system often needs to organise lists of spacecraft by various criteria, such as launch dates or mission priorities.

- (e) Given the following array representing spacecraft IDs: [5, 3, 8, 4, 2], demonstrate the process of sorting this array using the selection sort algorithm by showing the progression of the values in the array from the initial array to the final sorted array. (4 marks)

Description	Mark
For each of the four steps: (4 x 1 mark)	
Provides correct array after each step (description not required)	1
Total	4
Sample answer: - Find the smallest element in the array [5, 3, 8, 4, 2] which is 2, and swap it with the first element. - Array after Step 1: [2, 3, 8, 4, 5] - Find the smallest element in the remaining unsorted array [3, 8, 4, 5] which is 3, and swap it with the second element. - Array after Step 2: [2, 3, 8, 4, 5] - Find the smallest element in the remaining unsorted array [8, 4, 5] which is 4, and swap it with the third element. - Array after Step 3: [2, 3, 4, 8, 5] - Find the smallest element in the remaining unsorted array [8, 5] which is 5, and swap it with the fourth element. - Array after Step 4: [2, 3, 4, 5, 8]	
Accept other relevant answers	

Question 25**(36 marks)**

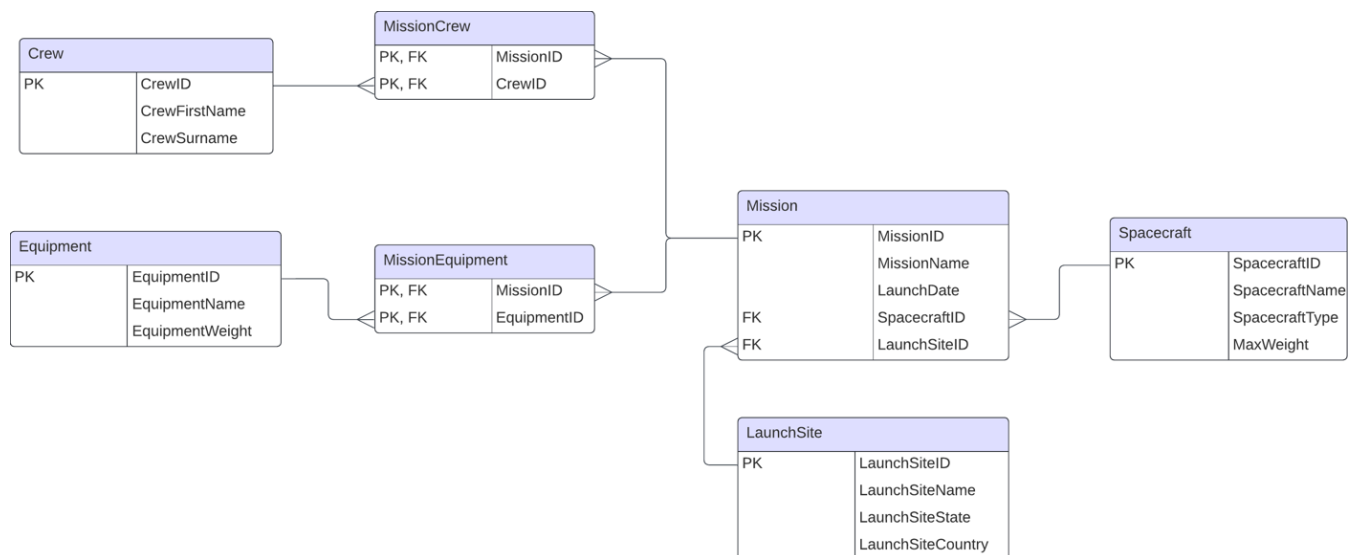
- (a) Your task is to normalise this data to Third Normal Form (3NF) and create an Entity Relationship Diagram (ERD). Draw your ERD in the space provided on the following page. (19 marks)

The ERD should clearly display all of the following:

- entities
- relationships (using crow's foot notation)
- primary keys
- foreign keys
- non-key fields

Additional Information

- Spacecraft can be involved in multiple missions, but each mission involves one spacecraft.
- Missions can have multiple crew members assigned, and each crew member can participate in multiple missions.
- Multiple pieces of equipment are required for each mission, and each piece of equipment can be used in multiple missions.
- Each launch site can be used for multiple missions.



Description	Mark
Entities	
All entities (7)	4
Most entities (5-6)	3
Some entities (3-4)	2
Few entities (1-2)	1
Crew MissionCrew Mission MissionEquipment Equipment Spacecraft LaunchSite	
Primary Keys	
All primary keys (7)	4
Most primary keys (5-6)	3
Some primary keys (3-4)	2
Few primary keys (1-2)	1
Crew - CrewID MissionCrew – MissionID/CrewID OR MissionCrewID Mission - MissionID MissionEquipment – MissionID/EquipmentID OR MissionEquipmentID Equipment – EquipmentID (also accept EquipmentName) Spacecraft - SpacecraftID LaunchSite – LaunchSiteID (also accept LaunchSiteName as this will be unique)	
Foreign Keys	
All foreign keys (6)	4
Most foreign keys (4-5)	3
Some foreign keys (2-3)	2
One foreign key (1)	1
MissionCrew – MissionID, CrewID Mission – SpacecraftID, LaunchSiteID MissionEquipment – MissionID, EquipmentID	
Cardinality	
All cardinalities (6)	4
Most cardinalities (4-5)	3
Some cardinalities (2-3)	2
One cardinality (1)	1
Non-key fields	
All non-key fields (5)	3
Most non-key fields (3-4)	2
Some non-key fields (1-2)	1
Crew – CrewFirstName, CrewSurname MissionCrew - None Mission – MissionName, LaunchDate MissionEquipment - None Equipment – EquipmentName (if not used as PK), EquipmentWeight Spacecraft – SpacecraftName, SpacecraftType, MaxWeight LaunchSite – LaunchSiteName (if not used as PK), LaunchSiteState, LaunchSiteCountry	
Total	19

Refer to the launch announcement poster on page 1 of the source booklet to assist you with this question.

(b) Write SQL queries to perform the following tasks:

- (i) You have been asked to use your SQL skills to retrieve some necessary data from the database to be used on a promotional poster. Information should be for the Mars Exploration mission (MissionID – 1151). (6 marks)

Description	Mark
SELECT Mission.MissionName, Spacecraft.SpacecraftName, Crew.CrewFirstName, Crew.CrewSurname, Mission.LaunchDate	1
FROM Mission	1
INNER JOIN Spacecraft ON Mission.SpacecraftID = Spacecraft.SpacecraftID	1
INNER JOIN MissionCrew ON Mission.MissionID = MissionCrew.MissionID	1
INNER JOIN Crew ON MissionCrew.CrewID = Crew.CrewID	1
WHERE Mission.MissionID = 1151	1
Total	6
Sample answer: SELECT Mission.MissionName, Spacecraft.SpacecraftName, Crew.CrewFirstName, Crew.CrewSurname, Mission.LaunchDate FROM Mission INNER JOIN Spacecraft ON Mission.SpacecraftID = Spacecraft.SpacecraftID INNER JOIN MissionCrew ON Mission.MissionID = MissionCrew.MissionID INNER JOIN Crew ON MissionCrew.CrewID = Crew.CrewID WHERE Mission.MissionID = 1151;	
Accept other relevant answers	

- (ii) Aisha Jackson (CrewID – 0311) has unfortunately fractured her ankle and is unable to take part in the mission. A new crew member, Mark Harmel (CrewID – 0405), will take her place. Write the SQL below to update the database with this information. (4 marks)

Description	Mark
UPDATE MissionCrew	1
SET CrewID = 0405	1
WHERE MissionID = 1151	1
AND CrewID = 0311;	1
Total	4
Sample answer: UPDATE MissionCrew SET CrewID = 0405 WHERE MissionID = 1151 AND CrewID = 0311;	
Accept other relevant answers	

(c) Data integrity is an important aspect of database design.

(i) Describe the concept of data integrity. (2 marks)

Description	Mark
Describes the concept of data integrity	2
Provides basic facts about the concept of data integrity	1
Total	2
Sample answers could include: Data integrity refers to the accuracy, consistency, and reliability of data stored in a database. It ensures that the data is correct and valid, free from errors, and remains consistent throughout its lifecycle, including during storage, retrieval, and processing. Maintaining data integrity is crucial for ensuring that the data can be trusted and is useful for making informed decisions.	
Accept other relevant answers	

(ii) Using an example from the SEA spacecraft operations data, describe **one** example of how referential integrity can be enforced in the database. (2 marks)

Description	Mark
Describes example of how referential integrity can be enforced in the database	2
Provides basic facts of how referential integrity can be enforced in the database	1
Total	2
Examples could include: - Ensuring that every CrewID in the MissionCrew table exists as a valid CrewID in the Crew table. - Maintaining that each SpacecraftID in the Mission table corresponds to an existing SpacecraftID in the Spacecraft table. - Guaranteeing that every EquipmentID in the MissionEquipment table is a valid EquipmentID in the Equipment table. - Making sure that LaunchSiteID in the Mission table matches an existing LaunchSiteID in the LaunchSite table. - Validating that MissionID in the MissionCrew and MissionEquipment tables refer to an existing MissionID in the Mission table.	
Accept other relevant answers	

- (d) Describe **one** privacy concern related to collecting data about individuals in the Space Exploration Agency's database and outline **one** strategy that could be implemented to protect personal information of crew members. (3 marks)

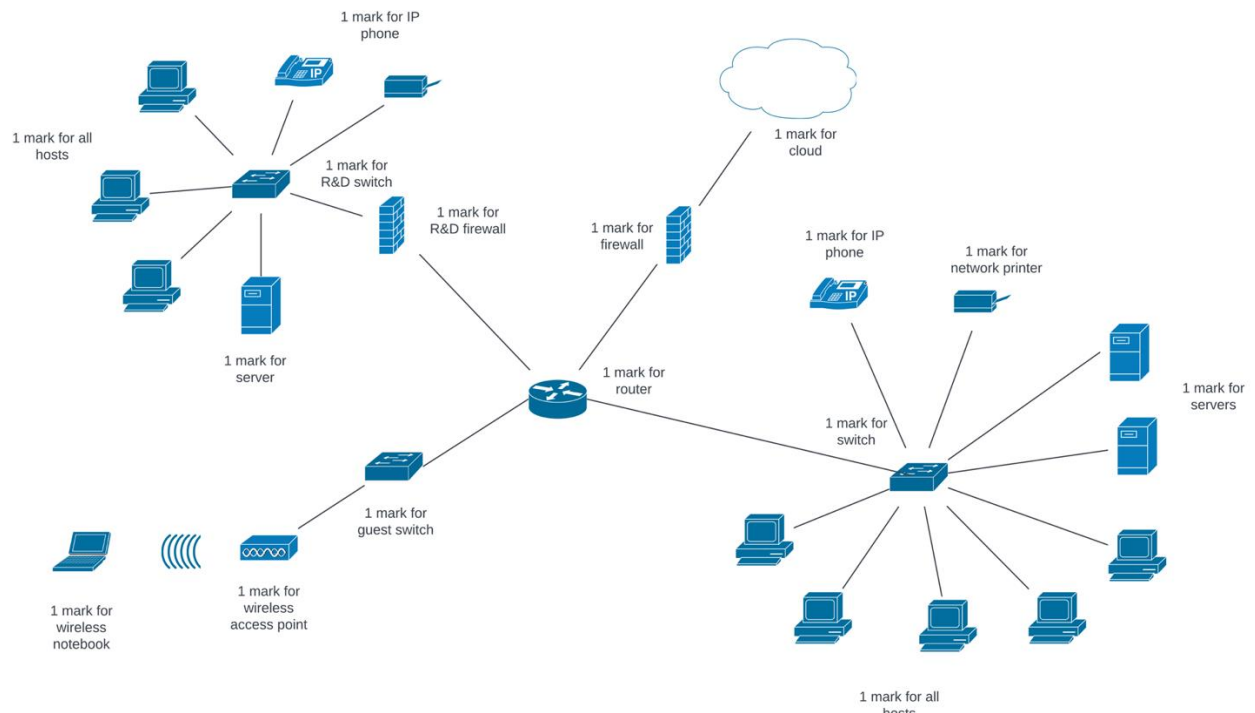
Description	Mark
Describes a relevant privacy concern from the spacecraft operations data	2
Provides superficial comment on a relevant privacy concern from the spacecraft operations data	1
Subtotal	2
Outlines a strategy to protect the personal information	1
Subtotal	1
Total	3
Sample answers could include: • The risk of unauthorised access to crew members' medical records, which could lead to sensitive health information being exposed or misused. ○ Protection Strategy: Implement access controls that ensure only authorised personnel can view medical records, and regularly audit access logs to detect and respond to any unauthorised access attempts. • The potential for data breaches that expose crew members' contact details, which could result in harassment or phishing attacks. ○ Protection Strategy: Use data anonymisation techniques to mask personal identifiers in non-essential contexts, ensuring that even if a breach occurs, the exposed data cannot be easily linked back to specific individuals. • The collection of crew members' location data can raise privacy issues, as constant tracking may lead to unwarranted surveillance and misuse of this information. ○ Protection Strategy: Implement strict policies and procedures for the collection, storage, and use of location data, ensuring that it is only accessed for legitimate purposes and with explicit consent from the individuals involved.	
Accept other relevant answers	

Question 26

(31 marks)

Refer to the information on page 2 of the source booklet to assist answer this question.

- (a) Draw a network diagram in the to represent the SEA Main Control Centre network infrastructure. You must use appropriate Cisco icons for all devices. (16 marks)



Description	Mark
Cloud	1
Firewall – correct icon used and connected to internet (cloud)	1
Router – correct icon used connected to firewall	1
Technician Subnet	
Switch – correct icon used and connected to router	1
5 x technician desktop computers – correct icons (workstation) used and connected to switch	1
Printer– correct icon used and connected to switch	1
2 x servers – correct icon used and connected to switch	1
IP phone – accept any icon – must be connected to switch	1
Guest Subnet	
Switch – correct icon used and connected to router	1
Wireless access point (WAP) - correct icon used and connected to switch	1
1 x wireless notebook computer - correct icon used and connected to wireless access spoint	1
Research & Development (R&D) Subnet	
Firewall – correct icon used and connected to router	1
Switch – correct icon used and connected to firewall	1
3 x technician desktop computers – correct icons (workstation) used and connected to switch	1
IP phone – correct icon used and connected to switch	1
Server – correct icon used and connected to switch	1
Printer– correct icon used and connected to switch	1
Total	16

While the decision to structure the local area network with multiple networks has been made, the IP addresses have not yet been allocated to each device.

Use the information in this table to answer the following question:

- Total IP address allocation: 192.168.15.0 with subnet mask 255.255.255.0
- You have decided that each of the subnets would require up to 60 devices.
 - First subnet – Mission Technician devices
 - Second subnet – Guest WiFi devices
 - Third subnet – Research & Development devices

As network administrator you always use a specific ruleset to determine which IP addresses are allocated to different devices on the network.

The following rules apply to ALL subnets (even those without all of the network devices):

Network Device	IP Address Allocation
Router port	First useable IP address
Servers	Second-fourth useable IP addresses
VOIP Phone	Second last useable IP address
Network printer	Last useable IP address
Workstations (wired or wireless)	All remaining useable IP addresses

(b) Provide IP addresses for the following:

- (i) Network address and subnet mask for the Research & Development subnet.

(1 mark)

Description	Mark
Network address: 192.168.15.128 Subnet mask: 255.255.255.192	1
Total	1

- (i) IP address and subnet mask for first connected wireless notebook on the Guest WiFi.

(1 mark)

Description	Mark
IP address: 192.168.15.69 Subnet mask: 255.255.255.192	1
Total	1

- (ii) IP address and subnet mask for the wired network printer used in the mission technician subnet. (1 mark)

Description	Mark
IP address: 192.168.15.62 Subnet mask: 255.255.255.192	1
Total	1

- (iii) The IP address and subnet mask for the router port connected to the Guest WiFi subnet. (1 mark)

Description	Mark
IP address: 192.168.15.65 Subnet mask: 255.255.255.192	1
Total	1

- (iv) The IP address and subnet mask for the first connected server in the Research & Development subnet. (1 mark)

Description	Mark
IP address: 192.168.15.130 Subnet mask: 255.255.255.192	1
Total	1

c) Default gateways are essential for network communication.

- i) Explain the role of a default gateway in a local area network (LAN). (3 marks)

Description	Mark
Explains the role of a default gateway in a LAN	3
Provides a basic explanation of the default gateway function with partial details	2
Provides superficial comment on the role of a default gateway	1
Total	3
Sample answer: The default gateway serves as an access point or IP router that a networked computer uses to send information to a computer in another network or the internet. It acts as a gate through which data passes to reach its destination outside the local network. Without a default gateway, devices on a local area network would not be able to communicate with devices outside their network, severely limiting network functionality.	
Accept other relevant answers	

- ii) Outline **one** similarity and **one** difference between the default gateway assigned for a workstation in the Research & Development (R&D) subnet and the default gateway assigned for a workstation in the Mission Technician subnet. (2 marks)

Description	Mark
Outlines appropriate similarity	1
Outlines appropriate difference	1
Total	2
Sample answer: Similarity: Identifies that they are the same piece of hardware (router) Difference: The IP address for each will be different as they will correspond to the router port assigned to the two different subnets	
Accept other relevant answers	

- d) One of the Research & Development engineers is attempting to connect to a website (bigastronomy.com.au) and has received the following message in her browser:



- i) Describe the most common problem to cause this error message. (2 marks)

Description	Mark
Describes the common issue causing the DNS error	2
Provides basic facts about the common issue causing the DNS error	1
Total	2
Sample answer: The most common problem causing this error message is a misconfiguration in the DNS settings. This can occur if the DNS server addresses are incorrect, the DNS cache is corrupted, or the DNS server itself is down or unresponsive.	
Accept other relevant answers	

- ii) Outline the purpose of the 'ping' command and describe **one** technique using the 'ping' command that could be used to test your theory from part (d)(i) of this question. (3 marks)

Description	Mark
Outlines the purpose of the ping command	1
Subtotal	3
Describes the way to use ping to check for DNS issue	2
Provides superficial comment regarding the use of ping to check for DNS issues	1
Subtotal	2
Total	3
Sample answer: The 'ping' command is used to test network connectivity by sending ICMP Echo Request packets to a destination and waiting for an ICMP Echo Reply, thereby measuring the time it takes for the round trip. Strategies could include: • Using 'ping' to Test DNS Issues: First, ping the DNS server to check its availability using the command ping <DNS server IP>. If the DNS server responds, this verifies that the server is reachable and the network path is intact. • Using 'ping' to Test DNS Issues: Next, ping the domain name bigastronomy.com.au to see if it resolves to an IP address using the command ping bigastronomy.com.au. If the ping fails, it indicates a DNS resolution issue, confirming the DNS settings may be misconfigured or the DNS server is unresponsive.	
Accept other relevant answers	

Question 27**(25 marks)**

SEA is aware that it needs to ensure that their premises and ICT infrastructure must be secure to protect sensitive and confidential data and ensure the integrity of its operations.

- (a) Identify **one** internal and **one** external potential cybersecurity threats that the SEA's software system could face. For each threat, outline **one** potential impact on the system and **one** measure that can be taken to mitigate it. (6 marks)

Description	Mark
For the internal and external cybersecurity threats	
Provides a valid threat	1
Outlines the potential impact on the system	1
Outlines an appropriate mitigation strategy	1
Total	6
Sample answers could include: Internal network threats: Lost or Stolen Devices - Threat: Lost or stolen devices. - Potential Impact on the System: Unauthorised access to sensitive data and network resources if the device is not properly secured, leading to data breaches and potential exploitation of network vulnerabilities. - Mitigation Measure: Implement device encryption, remote wipe capabilities, and ensure that all devices have strong authentication methods such as multi-factor authentication (MFA). Compromised Credentials - Threat: Compromised credentials. - Potential Impact on the System: Unauthorised access to the network and sensitive information, potentially allowing attackers to escalate privileges and cause significant harm to the system. - Mitigation Measure: Enforce strong password policies, implement MFA, and use security monitoring tools to detect and respond to unusual login activities. Misuse by Employees - Threat: Misuse by employees. - Potential Impact on the System: Intentional or unintentional misuse of network resources, leading to data leakage, security breaches, or disruption of services. - Mitigation Measure: Establish clear usage policies, conduct regular training on acceptable use and security practices, and monitor network activity to detect and prevent misuse. External network threats could include: SQL Injection Attack - Threat: SQL injection attack. - Potential Impact on the System: Unauthorised access to the database, allowing attackers to view, modify, or delete sensitive data, potentially leading to data breaches. - Mitigation Measure: Implement prepared statements and parameterised queries to prevent SQL injection vulnerabilities.	

See next page

DDoS Attack
- Threat: Distributed Denial of Service (DDoS) attack. - Potential Impact on the System: Overwhelming network resources with massive traffic, causing service outages and significant operational disruptions. - Mitigation Measure: Deploy DDoS protection solutions such as cloud-based traffic scrubbing and use a content delivery network (CDN) to absorb and mitigate attack traffic.
Phishing Attack
- Threat: Phishing attack. - Potential Impact on the System: Unauthorised access to sensitive data and network credentials, potentially leading to data breaches and financial loss. - Mitigation Measure: Conduct regular employee training on recognising phishing attempts and implement advanced email filtering solutions to detect and block malicious emails.
Other possible external network threats – backdoor, IP spoofing, man-in-the-middle, cross-site scripting, types of malware, physical network threats, zero day vulnerabilities.
Accept other relevant answers

- (b) Describe the role of Intrusion Prevention Systems (IPS) in SEA's network security and outline **one** example of how an IPS can prevent a cyber-attack on SEA's systems. (3 marks)

Description	Mark
Describes the role of Intrusion Prevention Systems	2
Provides basic facts about Intrusion Prevention Systems	1
Subtotal	2
Outlines an example	1
Subtotal	1
Total	3
Sample answer: An Intrusion Prevention System (IPS) monitors network traffic in real-time to detect and prevent malicious activities, enhancing SEA's network security by blocking attacks as they occur. Examples could include: - An IPS can prevent a SQL injection attack by identifying and blocking malicious SQL code patterns before they reach SEA's database, thereby protecting sensitive information. - An IPS can detect and block Distributed Denial of Service (DDoS) attacks by identifying unusual traffic patterns and filtering out malicious packets, ensuring the network remains available and responsive. - An IPS can stop malware infections by detecting and blocking traffic containing known malware signatures or behaviours, preventing the malware from infiltrating and compromising SEA's systems.	
Accept other relevant answers	

- (c) Explain how regular software updates and patches influence the security of the SEA's software system. (3 marks)

Description	Mark
Explains how regular software updates and patches influence the security of the software system	3
Describes the role and importance of updates and patches	2
Provides a superficial comment about updates and patches	1
Total	3
Sample answer: Regular software updates and patches are crucial for maintaining the security of the SEA's software system as they fix known vulnerabilities, improve functionality, and protect against emerging threats. By regularly updating software, SEA ensures that potential exploits are addressed promptly, reducing the risk of cyber-attacks. This proactive approach helps maintain the integrity, confidentiality, and availability of sensitive data and critical systems.	
Accept other relevant answers	

- (d) Explain, using a specific example, how user training could be used to mitigate against a potential cyber-attack at SEA. (3 marks)

Description	Mark
Explains how user training could be used to reduce a specific cyber-attack including an example	3
Describes how user training could be used to reduce a specific cyber-attack including an example	2
Provides a basic example of how user training could assist against a cyber-attack	1
Total	3
Sample answers could include: Phishing attack: User training can significantly mitigate phishing attacks. For instance, by educating employees on how to identify suspicious emails, such as those with unusual senders, urgent language, or unexpected attachments, users can avoid clicking on malicious links or downloading harmful files, thus preventing phishing attempts from compromising SEA's network. Malware infections: User training can help prevent malware infections. For example, training sessions that emphasise the importance of not downloading software from unverified sources can reduce the likelihood of employees inadvertently installing malware. This proactive approach helps protect SEA's systems from being compromised by malicious software.	
Accept other relevant answers	

- (e) SEA discovered that an external hacker had exploited vulnerabilities in their software system to install a backdoor. Explain the concept of a backdoor. (3 marks)

Description	Mark
Clearly explains the concept of a backdoor	3
Describes the purpose and function of a backdoor	2
Provides a basic definition of a backdoor	1
Total	3
Sample answer: A backdoor is a method by which an unauthorised user gains persistent access to a computer system or network by bypassing standard authentication processes. It is often installed through exploiting vulnerabilities in the software, allowing the attacker to enter the system unnoticed and maintain control over it without detection. Backdoors pose significant security risks as they enable continuous and unrestricted access, potentially leading to data breaches and further exploitation of the system.	
Accept other relevant answers	

- (f) SEA has experienced a number of cyber-attacks. Outline **two** mitigation strategies for each of the following attacks that SEA could implement to prevent such incidents in the future. (6 marks)
- (i) SEA discovered that an attacker had gained access to their network by guessing an employee's weak password.

Description	Mark
For each of the two strategies (2 x 1 mark)	
Outlines a relevant mitigation strategy	1
Total	2
Sample answers could include: • Enforce the use of complex passwords that include a mix of uppercase and lowercase letters, numbers, and special characters. • Require additional verification methods, such as a code sent to a mobile device, in addition to the password. • Regularly check for weak or compromised passwords and require users to update them.	
Accept other relevant answers	

- (ii) An SEA employee inadvertently downloaded a malicious email attachment that installed malware on their workstation, which then spread to other parts of the network.

Description	Mark
For each of the two strategies (2 x 1)	
Outlines a relevant mitigation strategy	1
Total	2
Sample answer: • Use email filtering solutions that can detect and block malicious attachments. • Educate employees on how to recognise and avoid phishing emails and malicious attachments. • Install antivirus and anti-malware software on all workstations to detect and prevent malware from being installed.	
Accept other relevant answers	

- (iii) SEA's software system was compromised when an attacker exploited a vulnerability in an outdated version of their web application.

Description	Mark
For each of the two strategies (2 x 1 mark)	
Outlines a relevant mitigation strategy	1
Total	2
Sample answer: • Ensure that all software, including web applications, is regularly updated and patched to fix known vulnerabilities. • Perform regular security assessments to identify and address vulnerabilities before they can be exploited. • Deploy web application firewalls (WAFs) to monitor and filter incoming traffic to web applications, protecting against common web-based attacks.	
Accept other relevant answers	

End of Examination